Version mechanisms and conditions for comparison

Aspect	BBRv1	BBRv2	BBRv3
Path model	Delivery-rate and minimum-RTT estimates	Delivery-rate and minimum-RTT estimates	Delivery-rate and minimum-RTT estimates
Probing	Eight-phase ProbeBW sequence	Refill / Up / Down / Cruise	Revised four-phase probing
Congestion feedback	Model-based rate control; TCP recovery remains active	Loss-based bounds; ECN response when enabled/eligible	Loss-based bounds; ECN response when eligible
Parameters	Reference Linux v5.4 design	Defaults depend on alpha revision and runtime module settings	Defaults depend on source revision and eligibility conditions
Performance	Evaluate under matched conditions	No universal fairness ranking	No universal fairness ranking

Mechanism overview; not a measured performance ranking. Exact parameters are reported in the manuscript version-parameter table.